

Introduction to Normal Distribution

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Outline of Notes

1) Univariate Normal:

- Distribution form
- Standard normal
- Probability calculations
- Affine transformations
- Parameter estimation

2) Bivariate Normal:

- Distribution form
- Probability calculations
- Affine transformations
- Conditional distributions

3) Multivariate Normal:

- Distribution form
- Probability calculations
- Affine transformations
- Conditional distributions
- Parameter estimation

4) Sampling Distributions:

- Univariate case
- Multivariate case

Univariate Normal

Normal Density Function (Univariate)

Given a variable $x \in \mathbb{R}$, the normal **probability density function** (pdf) is

$$\begin{aligned} f(x) &= \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \\ &= \frac{1}{\sigma\sqrt{2\pi}} \exp\left\{-\frac{(x-\mu)^2}{2\sigma^2}\right\} \end{aligned} \tag{1}$$

where

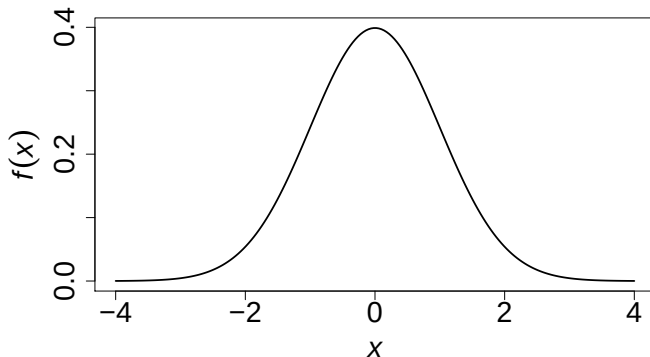
- $\mu \in \mathbb{R}$ is the **mean**
- $\sigma > 0$ is the **standard deviation** (σ^2 is the **variance**)
- $e \approx 2.71828$ is base of the natural logarithm

Write $X \sim N(\mu, \sigma^2)$ to denote that X follows a **normal distribution**.

Standard Normal Distribution

If $X \sim N(0, 1)$, then X follows a **standard normal distribution**:

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \quad (2)$$



Probabilities and Distribution Functions

Probabilities relate to the area under the pdf:

$$\begin{aligned}P(a \leq X \leq b) &= \int_a^b f(x)dx \\ &= F(b) - F(a)\end{aligned}\tag{3}$$

where

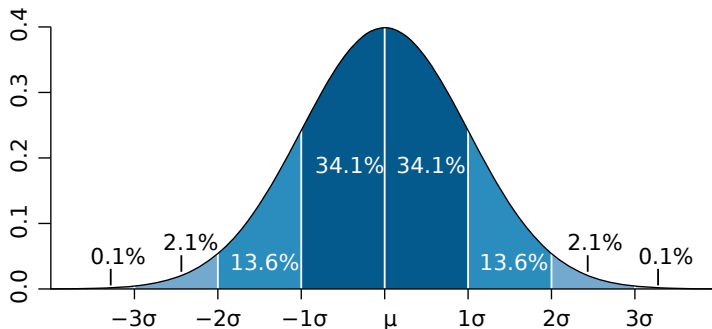
$$F(x) = \int_{-\infty}^x f(u)du\tag{4}$$

is the **cumulative distribution function** (cdf).

Note: $F(x) = P(X \leq x) \implies 0 \leq F(x) \leq 1$

Normal Probabilities

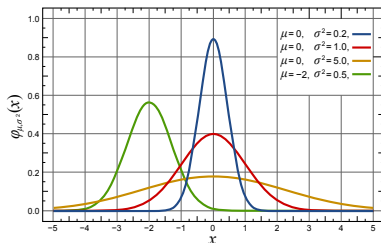
Helpful figure of normal probabilities:



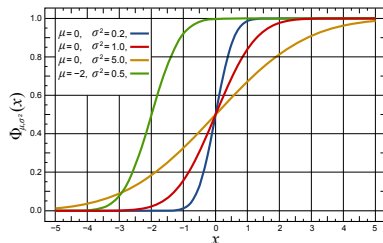
From http://en.wikipedia.org/wiki/File:Standard_deviation_diagram.svg

Normal Distribution Functions (Univariate)

Helpful figures of normal pdfs and cdfs:



http://en.wikipedia.org/wiki/File:Normal_Distribution_PDF.svg



http://en.wikipedia.org/wiki/File:Normal_Distribution_CDF.svg

Note that the cdf has an elongated “S” shape, referred to as an **ogive**.

Affine Transformations of Normal (Univariate)

Suppose that $X \sim N(\mu, \sigma^2)$ and $a, b \in \mathbb{R}$ with $a \neq 0$.

If we define $Y = aX + b$, then $Y \sim N(a\mu + b, a^2\sigma^2)$.

Suppose that $X \sim N(1, 2)$. Determine the distributions of...

- $Y = X + 3$
- $Y = 2X + 3$
- $Y = 3X + 2$

Affine Transformations of Normal (Univariate)

Suppose that $X \sim N(\mu, \sigma^2)$ and $a, b \in \mathbb{R}$ with $a \neq 0$.

If we define $Y = aX + b$, then $Y \sim N(a\mu + b, a^2\sigma^2)$.

Suppose that $X \sim N(1, 2)$. Determine the distributions of . . .

- $Y = X + 3 \implies Y \sim N(1(1) + 3, 1^2(2)) \equiv N(4, 2)$
- $Y = 2X + 3 \implies Y \sim N(2(1) + 3, 2^2(2)) \equiv N(5, 8)$
- $Y = 3X + 2 \implies Y \sim N(3(1) + 2, 3^2(2)) \equiv N(5, 18)$

Likelihood Function

Suppose that $\mathbf{x} = (x_1, \dots, x_n)$ is an iid sample of data from a normal distribution with mean μ and variance σ^2 , i.e., $x_i \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$.

The likelihood function for the parameters (given the data) has the form

$$L(\mu, \sigma^2 | \mathbf{x}) = \prod_{i=1}^n f(x_i) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(x_i - \mu)^2}{2\sigma^2} \right\}$$

and the log-likelihood function is given by

$$LL(\mu, \sigma^2 | \mathbf{x}) = \sum_{i=1}^n \log(f(x_i)) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

Maximum Likelihood Estimate of the Mean

The MLE of the mean is the value of μ that minimizes

$$\sum_{i=1}^n (x_i - \mu)^2 = \sum_{i=1}^n x_i^2 - 2n\bar{x}\mu + n\mu^2$$

where $\bar{x} = (1/n) \sum_{i=1}^n x_i$ is the sample mean.

Taking the derivative with respect to μ we find that

$$\frac{\partial \sum_{i=1}^n (x_i - \mu)^2}{\partial \mu} = -2n\bar{x} + 2n\mu \quad \longleftrightarrow \quad \bar{x} = \hat{\mu}$$

i.e., the sample mean $\bar{x}$ is the MLE of the population mean μ .

Maximum Likelihood Estimate of the Variance

The MLE of the variance is the value of σ^2 that minimizes

$$\frac{n}{2} \log(\sigma^2) + \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \hat{\mu})^2 = \frac{n}{2} \log(\sigma^2) + \frac{\sum_{i=1}^n x_i^2}{2\sigma^2} - \frac{n\bar{x}^2}{2\sigma^2}$$

where $\bar{x} = (1/n) \sum_{i=1}^n x_i$ is the sample mean.

Taking the derivative with respect to σ^2 we find that

$$\frac{\frac{n}{2} \log(\sigma^2) + \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \hat{\mu})^2}{\partial \sigma^2} = \frac{n}{2\sigma^2} - \frac{1}{2\sigma^4} \sum_{i=1}^n (x_i - \hat{\mu})^2$$

which implies that the sample variance $\hat{\sigma}^2 = (1/n) \sum_{i=1}^n (x_i - \bar{x})^2$ is the MLE of the population variance σ^2 .

Bivariate Normal

Normal Density Function (Bivariate)

Given two variables $x, y \in \mathbb{R}$, the **bivariate normal** pdf is

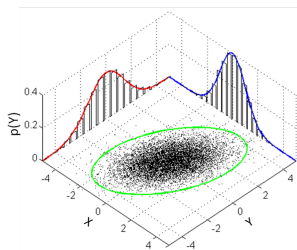
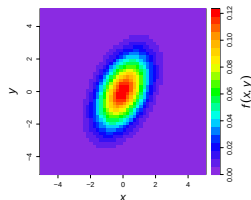
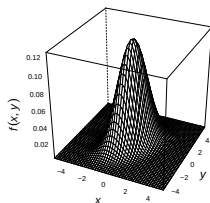
$$f(x, y) = \frac{\exp \left\{ -\frac{1}{2(1-\rho^2)} \left[\frac{(x-\mu_x)^2}{\sigma_x^2} + \frac{(y-\mu_y)^2}{\sigma_y^2} - \frac{2\rho(x-\mu_x)(y-\mu_y)}{\sigma_x\sigma_y} \right] \right\}}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \quad (5)$$

where

- $\mu_x \in \mathbb{R}$ and $\mu_y \in \mathbb{R}$ are the marginal means
- $\sigma_x \in \mathbb{R}^+$ and $\sigma_y \in \mathbb{R}^+$ are the marginal standard deviations
- $0 \leq |\rho| < 1$ is the correlation coefficient

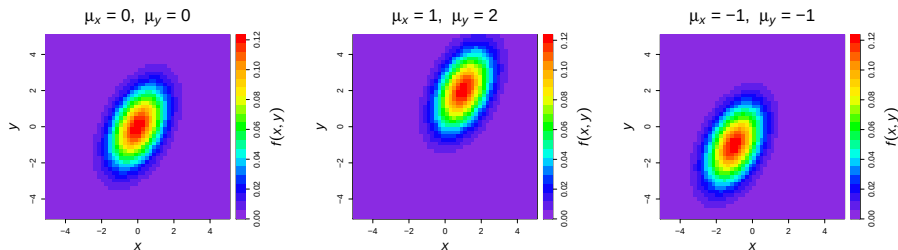
X and Y are marginally normal: $X \sim N(\mu_x, \sigma_x^2)$ and $Y \sim N(\mu_y, \sigma_y^2)$

Example: $\mu_x = \mu_y = 0$, $\sigma_x^2 = 1$, $\sigma_y^2 = 2$, $\rho = 0.6/\sqrt{2}$



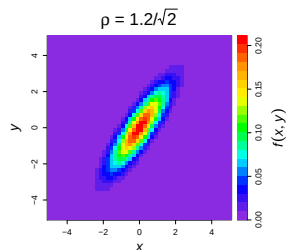
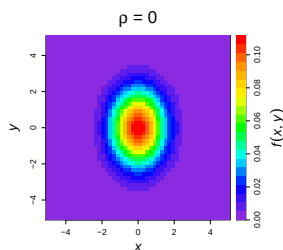
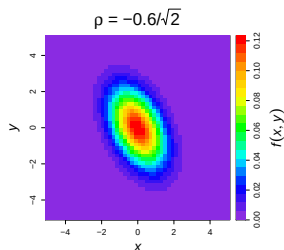
<http://en.wikipedia.org/wiki/File:MultivariateNormal.png>

Example: Different Means



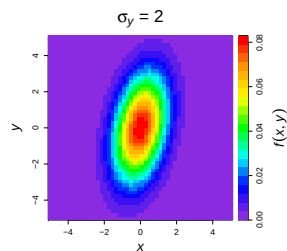
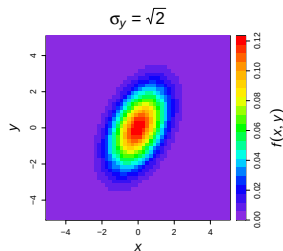
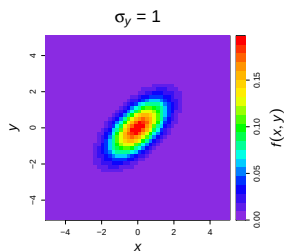
Note: for all three plots $\sigma_x^2 = 1$, $\sigma_y^2 = 2$, and $\rho = 0.6/\sqrt{2}$.

Example: Different Correlations



Note: for all three plots $\mu_x = \mu_y = 0$, $\sigma_x^2 = 1$, and $\sigma_y^2 = 2$.

Example: Different Variances



Note: for all three plots $\mu_x = \mu_y = 0$, $\sigma_x^2 = 1$, and $\rho = 0.6/(\sigma_x\sigma_y)$.

Probabilities and Multiple Integration

Probabilities still relate to the area under the pdf:

$$P([a_x \leq X \leq b_x] \text{ and } [a_y \leq Y \leq b_y]) = \int_{a_x}^{b_x} \int_{a_y}^{b_y} f(x, y) dy dx \quad (6)$$

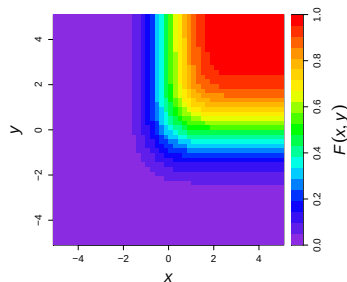
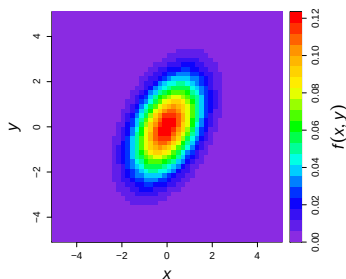
where $\int \int f(x, y) dy dx$ denotes the multiple integral of the pdf $f(x, y)$.

Defining $\mathbf{z} = (x, y)$, we can still define the cdf:

$$\begin{aligned} F(\mathbf{z}) &= P(X \leq x \text{ and } Y \leq y) \\ &= \int_{-\infty}^x \int_{-\infty}^y f(u, v) dv du \end{aligned} \quad (7)$$

Normal Distribution Functions (Bivariate)

Helpful figures of bivariate normal pdf and cdf:



Note: $\mu_x = \mu_y = 0$, $\sigma_x^2 = 1$, $\sigma_y^2 = 2$, and $\rho = 0.6/\sqrt{2}$

Note that the cdf still has an ogive shape (now in two-dimensions).

Affine Transformations of Normal (Bivariate)

Given $\mathbf{z} = (x, y)'$, suppose that $\mathbf{z} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ where

- $\boldsymbol{\mu} = (\mu_x, \mu_y)'$ is the 2×1 mean vector
- $\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_x^2 & \rho\sigma_x\sigma_y \\ \rho\sigma_x\sigma_y & \sigma_y^2 \end{pmatrix}$ is the 2×2 covariance matrix

Let $\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$ with $\mathbf{A} \neq \mathbf{0}_{2 \times 2} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$.

If we define $\mathbf{w} = \mathbf{A}\mathbf{z} + \mathbf{b}$, then $\mathbf{w} \sim N(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}')$.

Conditional Normal (Bivariate)

The **conditional distribution** of a variable Y given $X = x$ is

$$f_{Y|X}(y|X = x) = \frac{f_{XY}(x, y)}{f_X(x)} \quad (8)$$

where

- $f_{XY}(x, y)$ is the **joint pdf** of X and Y
- $f_X(x)$ is the **marginal pdf** of X

In the bivariate normal case, we have that

$$Y|X \sim N(\mu_*, \sigma_*^2) \quad (9)$$

where $\mu_* = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(x - \mu_X)$ and $\sigma_*^2 = \sigma_Y^2(1 - \rho^2)$

Derivation of Conditional Normal

To prove Equation (9), simply write out the definition and simplify:

$$\begin{aligned}
 f_{Y|X}(y|X=x) &= \frac{f_{XY}(x,y)}{f_X(x)} \\
 &= \frac{\exp \left\{ -\frac{1}{2(1-\rho^2)} \left[\frac{(x-\mu_X)^2}{\sigma_X^2} + \frac{(y-\mu_Y)^2}{\sigma_Y^2} - \frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} \right] \right\}}{\exp \left\{ -\frac{(x-\mu_X)^2}{2\sigma_X^2} \right\}} \cdot \frac{1}{\left(2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2} \right)} \\
 &= \frac{\exp \left\{ -\frac{1}{2(1-\rho^2)} \left[\frac{(x-\mu_X)^2}{\sigma_X^2} + \frac{(y-\mu_Y)^2}{\sigma_Y^2} - \frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} \right] + \frac{(x-\mu_X)^2}{2\sigma_X^2} \right\}}{\sqrt{2\pi}\sigma_Y\sqrt{1-\rho^2}} \\
 &= \frac{\exp \left\{ -\frac{1}{2\sigma_Y^2(1-\rho^2)} \left[\rho^2 \frac{\sigma_Y^2}{\sigma_X^2} (x-\mu_X)^2 + (y-\mu_Y)^2 - 2\rho \frac{\sigma_Y}{\sigma_X} (x-\mu_X)(y-\mu_Y) \right] \right\}}{\sqrt{2\pi}\sigma_Y\sqrt{1-\rho^2}} \\
 &= \frac{\exp \left\{ -\frac{1}{2\sigma_Y^2(1-\rho^2)} \left[y - \mu_Y - \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X) \right]^2 \right\}}{\sqrt{2\pi}\sigma_Y\sqrt{1-\rho^2}}
 \end{aligned}$$

which completes the proof.

Statistical Independence for Bivariate Normal

Two variables X and Y are **statistically independent** if

$$f_{XY}(x, y) = f_X(x)f_Y(y) \quad (10)$$

where $f_{XY}(x, y)$ is joint pdf, and $f_X(x)$ and $f_Y(y)$ are marginals pdfs.

Note that if X and Y are independent, then

$$f_{Y|X}(y|X = x) = \frac{f_{XY}(x, y)}{f_X(x)} = \frac{f_X(x)f_Y(y)}{f_X(x)} = f_Y(y) \quad (11)$$

so conditioning on $X = x$ does not change the distribution of Y .

If X and Y are bivariate normal, what is the necessary and sufficient condition for X and Y to be independent? Hint: see Equation (9)

Example #1

A statistics class takes two exams X (Exam 1) and Y (Exam 2) where the scores follow a bivariate normal distribution with parameters:

- $\mu_X = 70$ and $\mu_Y = 60$ are the marginal means
- $\sigma_X = 10$ and $\sigma_Y = 15$ are the marginal standard deviations
- $\rho = 0.6$ is the correlation coefficient

Suppose we select a student at random. What is the probability that. . .

- (a) the student scores over 75 on Exam 2?
- (b) the student scores over 75 on Exam 2, given that the student scored $X = 80$ on Exam 1?
- (c) the sum of his/her Exam 1 and Exam 2 scores is over 150?
- (d) the student did better on Exam 1 than Exam 2?
- (e) $P(5X - 4Y > 150)$?

Example #1: Part (a)

Answer for 1(a):

Note that $Y \sim N(60, 15^2)$, so the probability that the student scores over 75 on Exam 2 is

$$\begin{aligned}P(Y > 75) &= P\left(Z > \frac{75 - 60}{15}\right) \\&= P(Z > 1) \\&= 1 - P(Z < 1) \\&= 1 - \Phi(1) \\&= 1 - 0.8413447 \\&= 0.1586553\end{aligned}$$

where $\Phi(x) = \int_{-\infty}^x f(z)dz$ with $f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ denoting the standard normal pdf (see R code for use of `pnorm` to calculate this quantity).

Example #1: Part (b)

Answer for 1(b):

Note that $(Y|X = 80) \sim N(\mu_*, \sigma_*^2)$ where

$$\mu_* = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(x - \mu_X) = 60 + (0.6)(15/10)(80 - 70) = 69$$

$$\sigma_*^2 = \sigma_Y^2(1 - \rho^2) = 15^2(1 - 0.6^2) = 144$$

If a student scored $X = 80$ on Exam 1, the probability that the student scores over 75 on Exam 2 is

$$\begin{aligned} P(Y > 75|X = 80) &= P\left(Z > \frac{75 - 69}{12}\right) \\ &= P(Z > 0.5) \\ &= 1 - \Phi(0.5) \\ &= 1 - 0.6914625 \\ &= 0.3085375 \end{aligned}$$

Example #1: Part (c)

Answer for 1(c):

Note that $(X + Y) \sim N(\mu_*, \sigma_*^2)$ where

$$\mu_* = \mu_X + \mu_Y = 70 + 60 = 130$$

$$\sigma_*^2 = \sigma_X^2 + \sigma_Y^2 + 2\rho\sigma_X\sigma_Y = 10^2 + 15^2 + 2(0.6)(10)(15) = 505$$

The probability that the sum of Exam 1 and Exam 2 is above 150 is

$$\begin{aligned} P(X + Y > 150) &= P\left(Z > \frac{150 - 130}{\sqrt{505}}\right) \\ &= P(Z > 0.8899883) \\ &= 1 - \Phi(0.8899883) \\ &= 1 - 0.8132639 \\ &= 0.1867361 \end{aligned}$$

Example #1: Part (d)

Answer for 1(d):

Note that $(X - Y) \sim N(\mu_*, \sigma_*^2)$ where

$$\mu_* = \mu_X - \mu_Y = 70 - 60 = 10$$

$$\sigma_*^2 = \sigma_X^2 + \sigma_Y^2 - 2\rho\sigma_X\sigma_Y = 10^2 + 15^2 - 2(0.6)(10)(15) = 145$$

The probability that the student did better on Exam 1 than Exam 2 is

$$\begin{aligned} P(X > Y) &= P(X - Y > 0) \\ &= P\left(Z > \frac{0 - 10}{\sqrt{145}}\right) \\ &= P(Z > -0.8304548) \\ &= 1 - \Phi(-0.8304548) \\ &= 1 - 0.2031408 \\ &= 0.7968592 \end{aligned}$$

Example #1: Part (e)

Answer for 1(e):

Note that $(5X - 4Y) \sim N(\mu_*, \sigma_*^2)$ where

$$\mu_* = 5\mu_X - 4\mu_Y = 5(70) - 4(60) = 110$$

$$\begin{aligned}\sigma_*^2 &= 5^2\sigma_X^2 + (-4)^2\sigma_Y^2 + 2(5)(-4)\rho\sigma_X\sigma_Y = \\ &25(10^2) + 16(15^2) - 2(20)(0.6)(10)(15) = 2500\end{aligned}$$

Thus, the needed probability can be obtained using

$$\begin{aligned}P(5X - 4Y > 150) &= P\left(Z > \frac{150 - 110}{\sqrt{2500}}\right) \\ &= P(Z > 0.8) \\ &= 1 - \Phi(0.8) \\ &= 1 - 0.7881446 \\ &= 0.2118554\end{aligned}$$

Example #1: R Code

```
# Example 1a
> pnorm(1,lower=F)
[1] 0.1586553
> pnorm(75,mean=60,sd=15,lower=F)
[1] 0.1586553
```

```
# Example 1b
> pnorm(0.5,lower=F)
[1] 0.3085375
> pnorm(75,mean=69,sd=12,lower=F)
[1] 0.3085375
```

```
# Example 1c
> pnorm(20/sqrt(505),lower=F)
[1] 0.1867361
> pnorm(150,mean=130,sd=sqrt(505),lower=F)
[1] 0.1867361
```

```
# Example 1d
> pnorm(-10/sqrt(145),lower=F)
[1] 0.7968592
> pnorm(0,mean=10,sd=sqrt(145),lower=F)
[1] 0.7968592
```

```
# Example 1e
> pnorm(0.8,lower=F)
[1] 0.2118554
> pnorm(150,mean=110,sd=50,lower=F)
[1] 0.2118554
```

Example #1: Python Code

```

❶ from math import sqrt
   from scipy.stats import norm

   # In R: pnorm(x, lower=F)  <=>  in Python: norm.sf(x) = 1 - norm.cdf(x)

   # Example 1a
   ans_1a_z = norm.sf(1) # P(Z > 1)
   ans_1a_y = norm.sf(75, loc=60, scale=15) # P(Y > 75), Y~N(60,15^2)
   print("1a:", ans_1a_z, ans_1a_y)

   # Example 1b
   ans_1b_z = norm.sf(0.5) # P(Z > 0.5)
   ans_1b_y = norm.sf(75, loc=69, scale=12) # P(Y > 75 | X=80), Y|X~N(69,12^2)
   print("1b:", ans_1b_z, ans_1b_y)

   # Example 1c
   ans_1c_z = norm.sf(20 / sqrt(505)) # P(Z > 20/sqrt(505))
   ans_1c_sum = norm.sf(150, loc=130, scale=sqrt(505)) # P(X+Y > 150), X+Y~N(130,505)
   print("1c:", ans_1c_z, ans_1c_sum)

   # Example 1d
   ans_1d_z = norm.sf(-10 / sqrt(145)) # P(Z > -10/sqrt(145))
   ans_1d_diff = norm.sf(0, loc=10, scale=sqrt(145)) # P(X-Y > 0), X-Y~N(10,145)
   print("1d:", ans_1d_z, ans_1d_diff)

   # Example 1e
   ans_1e_z = norm.sf(0.8) # P(Z > 0.8)
   ans_1e_lin = norm.sf(150, loc=110, scale=50) # P(5X-4Y > 150), 5X-4Y~N(110,2500)
   print("1e:", ans_1e_z, ans_1e_lin)

... 1a: 0.15865525393145707 0.15865525393145707
    1b: 0.3085375387259869 0.3085375387259869
    1c: 0.1867360791433823 0.1867360791433823
    1d: 0.7968591529412239 0.7968591529412239
    1e: 0.2118553985833967 0.2118553985833967

```



Multivariate Normal

Normal Density Function (Multivariate)

Given $\mathbf{x} = (x_1, \dots, x_p)'$ with $x_j \in \mathbb{R} \forall j$, the **multivariate normal** pdf is

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{p/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left\{ -\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right\} \quad (12)$$

where

- $\boldsymbol{\mu} = (\mu_1, \dots, \mu_p)'$ is the $p \times 1$ **mean vector**

- $\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1p} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{p1} & \sigma_{p2} & \cdots & \sigma_{pp} \end{pmatrix}$ is the $p \times p$ **covariance matrix**

Write $\mathbf{x} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ or $\mathbf{x} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ to denote $\mathbf{x}$ is multivariate normal.

Some Multivariate Normal Properties

The mean and covariance parameters have the following restrictions:

- $\mu_j \in \mathbb{R}$ for all j
- $\sigma_{jj} > 0$ for all j
- $\sigma_{ij} = \rho_{ij}\sqrt{\sigma_{ii}\sigma_{jj}}$ where ρ_{ij} is correlation between X_i and X_j
- $\sigma_{ij}^2 \leq \sigma_{ii}\sigma_{jj}$ for any $i, j \in \{1, \dots, p\}$ (Cauchy-Schwarz)

Σ is assumed to be positive definite so that Σ^{-1} exists.

Marginals are normal: $X_j \sim N(\mu_j, \sigma_{jj})$ for all $j \in \{1, \dots, p\}$.

Multivariate Normal Probabilities

Probabilities still relate to the area under the pdf:

$$P(a_j \leq X_j \leq b_j \forall j) = \int_{a_1}^{b_1} \cdots \int_{a_p}^{b_p} f(\mathbf{x}) dx_p \cdots dx_1 \quad (13)$$

where $\int \cdots \int f(\mathbf{x}) dx_p \cdots dx_1$ denotes the multiple integral $f(\mathbf{x})$.

We can still define the cdf of $\mathbf{x} = (x_1, \dots, x_p)'$

$$\begin{aligned} F(\mathbf{x}) &= P(X_j \leq x_j \forall j) \\ &= \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_p} f(\mathbf{u}) du_p \cdots du_1 \end{aligned} \quad (14)$$

Affine Transformations of Normal (Multivariate)

Suppose that $\mathbf{x} = (x_1, \dots, x_p)'$ and that $\mathbf{x} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ where

- $\boldsymbol{\mu} = \{\mu_j\}_{p \times 1}$ is the mean vector
- $\boldsymbol{\Sigma} = \{\sigma_{ij}\}_{p \times p}$ is the covariance matrix

Let $\mathbf{A} = \{a_{ij}\}_{n \times p}$ and $\mathbf{b} = \{b_i\}_{n \times 1}$ with $\mathbf{A} \neq \mathbf{0}_{n \times p}$.

If we define $\mathbf{w} = \mathbf{Ax} + \mathbf{b}$, then $\mathbf{w} \sim N(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}')$.

Note: linear combinations of normal variables are normally distributed.

Multivariate Conditional Distributions

Given variables $\mathbf{x} = (x_1, \dots, x_p)'$ and $\mathbf{y} = (y_1, \dots, y_q)'$, we have

$$f_{Y|X}(\mathbf{y}|X = \mathbf{x}) = \frac{f_{XY}(\mathbf{x}, \mathbf{y})}{f_X(\mathbf{x})} \quad (15)$$

where

- $f_{Y|X}(\mathbf{y}|X = \mathbf{x})$ is the conditional distribution of $\mathbf{y}$ given $\mathbf{x}$
- $f_{XY}(\mathbf{x}, \mathbf{y})$ is the joint pdf of $\mathbf{x}$ and $\mathbf{y}$
- $f_X(\mathbf{x})$ is the marginal pdf of $\mathbf{x}$

Conditional Normal (Multivariate)

Suppose that $\mathbf{z} \sim \mathbf{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ where

- $\mathbf{z} = (\mathbf{x}', \mathbf{y}')' = (x_1, \dots, x_p, y_1, \dots, y_q)'$

- $\boldsymbol{\mu} = (\boldsymbol{\mu}'_x, \boldsymbol{\mu}'_y)' = (\mu_{1x}, \dots, \mu_{px}, \mu_{1y}, \dots, \mu_{qy})'$

Note: $\boldsymbol{\mu}_x$ is mean vector of $\mathbf{x}$, and $\boldsymbol{\mu}_y$ is mean vector of $\mathbf{y}$

- $\boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{xx} & \boldsymbol{\Sigma}_{xy} \\ \boldsymbol{\Sigma}'_{xy} & \boldsymbol{\Sigma}_{yy} \end{pmatrix}$ where $(\boldsymbol{\Sigma}_{xx})_{p \times p}$, $(\boldsymbol{\Sigma}_{yy})_{q \times q}$, and $(\boldsymbol{\Sigma}_{xy})_{p \times q}$,

Note: $\boldsymbol{\Sigma}_{xx}$ is covariance matrix of $\mathbf{x}$, $\boldsymbol{\Sigma}_{yy}$ is covariance matrix of $\mathbf{y}$, and $\boldsymbol{\Sigma}_{xy}$ is covariance matrix of $\mathbf{x}$ and $\mathbf{y}$

In the multivariate normal case, we have that

$$\mathbf{y}|\mathbf{x} \sim \mathbf{N}(\boldsymbol{\mu}_*, \boldsymbol{\Sigma}_*) \quad (16)$$

where $\boldsymbol{\mu}_* = \boldsymbol{\mu}_y + \boldsymbol{\Sigma}'_{xy} \boldsymbol{\Sigma}_{xx}^{-1} (\mathbf{x} - \boldsymbol{\mu}_x)$ and $\boldsymbol{\Sigma}_* = \boldsymbol{\Sigma}_{yy} - \boldsymbol{\Sigma}'_{xy} \boldsymbol{\Sigma}_{xx}^{-1} \boldsymbol{\Sigma}_{xy}$

Statistical Independence for Multivariate Normal

Using Equation (16), we have that

$$\mathbf{y}|\mathbf{x} \sim N(\boldsymbol{\mu}_*, \boldsymbol{\Sigma}_*) \equiv N(\boldsymbol{\mu}_y, \boldsymbol{\Sigma}_{yy}) \quad (17)$$

if and only if $\boldsymbol{\Sigma}_{xy} = \mathbf{0}_{p \times q}$ (a matrix of zeros).

Note that $\boldsymbol{\Sigma}_{xy} = \mathbf{0}_{p \times q}$ implies that the p elements of $\mathbf{x}$ are uncorrelated with the q elements of $\mathbf{y}$.

- For multivariate normal variables: uncorrelated $\rightarrow$ independent
- For non-normal variables: uncorrelated $\nrightarrow$ independent

Example #2

Each Delicious Candy Company store makes 3 size candy bars: regular (X_1), fun size (X_2), and big size (X_3).

Assume the weight (in ounces) of the candy bars (X_1, X_2, X_3) follow a multivariate normal distribution with parameters:

$$\bullet \mu = \begin{pmatrix} 5 \\ 3 \\ 7 \end{pmatrix} \quad \text{and} \quad \Sigma = \begin{pmatrix} 4 & -1 & 0 \\ -1 & 4 & 2 \\ 0 & 2 & 9 \end{pmatrix}$$

Suppose we select a store at random. What is the probability that...

- (a) the weight of a regular candy bar is greater than 8 oz?
- (b) the weight of a regular candy bar is greater than 8 oz, given that the fun size bar weighs 1 oz and the big size bar weighs 10 oz?
- (c) $P(4X_1 - 3X_2 + 5X_3 < 63)$?

Example #2: Part (a)

Answer for 2(a):

Note that $X_1 \sim N(5, 4)$

So, the probability that the regular bar is more than 8 oz is

$$\begin{aligned} P(X_1 > 8) &= P\left(Z > \frac{8-5}{2}\right) \\ &= P(Z > 1.5) \\ &= 1 - \Phi(1.5) \\ &= 1 - 0.9331928 \\ &= 0.0668072 \end{aligned}$$

Example #2: Part (b)

Answer for 2(b):

$(X_1|X_2 = 1, X_3 = 10)$ is normally distributed, see Equation (16).

The conditional mean of $(X_1|X_2 = 1, X_3 = 10)$ is given by

$$\begin{aligned}\mu_* &= \mu_{X_1} + \boldsymbol{\Sigma}'_{12} \boldsymbol{\Sigma}_{22}^{-1} (\tilde{\mathbf{x}} - \tilde{\boldsymbol{\mu}}) \\ &= 5 + (-1 \quad 0) \begin{pmatrix} 4 & 2 \\ 2 & 9 \end{pmatrix}^{-1} \begin{pmatrix} 1 - 3 \\ 10 - 7 \end{pmatrix} \\ &= 5 + (-1 \quad 0) \frac{1}{32} \begin{pmatrix} 9 & -2 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} -2 \\ 3 \end{pmatrix} \\ &= 5 + 24/32 \\ &= 5.75\end{aligned}$$

Example #2: Part (b) continued

Answer for 2(b) continued:

The conditional variance of $(X_1|X_2 = 1, X_3 = 10)$ is given by

$$\begin{aligned}\sigma_*^2 &= \sigma_{X_1}^2 - \boldsymbol{\Sigma}'_{12} \boldsymbol{\Sigma}_{22}^{-1} \boldsymbol{\Sigma}_{12} \\ &= 4 - (-1 \quad 0) \begin{pmatrix} 4 & 2 \\ 2 & 9 \end{pmatrix}^{-1} \begin{pmatrix} -1 \\ 0 \end{pmatrix} \\ &= 4 - (-1 \quad 0) \frac{1}{32} \begin{pmatrix} 9 & -2 \\ -2 & 4 \end{pmatrix} \begin{pmatrix} -1 \\ 0 \end{pmatrix} \\ &= 4 - 9/32 \\ &= 3.71875\end{aligned}$$

Example #2: Part (b) continued

Answer for 2(b) continued:

So, if the fun size bar weighs 1 oz and the big size bar weighs 10 oz, the probability that the regular bar is more than 8 oz is

$$\begin{aligned}P(X_1 > 8 | X_2 = 1, X_3 = 10) &= P\left(Z > \frac{8 - 5.75}{\sqrt{3.71875}}\right) \\&= P(Z > 1.166767) \\&= 1 - \Phi(1.166767) \\&= 1 - 0.8783477 \\&= 0.1216523\end{aligned}$$

Example #2: Part (c)

Answer for 2(c):

$(4X_1 - 3X_2 + 5X_3)$ is normally distributed.

The expectation of $(4X_1 - 3X_2 + 5X_3)$ is given by

$$\begin{aligned}\mu_* &= 4\mu_{X_1} - 3\mu_{X_2} + 5\mu_{X_3} \\ &= 4(5) - 3(3) + 5(7) \\ &= 46\end{aligned}$$

Example #2: Part (c) continued

Answer for 2(c) continued:

The variance of $(4X_1 - 3X_2 + 5X_3)$ is given by

$$\begin{aligned}\sigma_*^2 &= (4 \quad -3 \quad 5) \Sigma \begin{pmatrix} 4 \\ -3 \\ 5 \end{pmatrix} \\ &= (4 \quad -3 \quad 5) \begin{pmatrix} 4 & -1 & 0 \\ -1 & 4 & 2 \\ 0 & 2 & 9 \end{pmatrix} \begin{pmatrix} 4 \\ -3 \\ 5 \end{pmatrix} \\ &= (4 \quad -3 \quad 5) \begin{pmatrix} 19 \\ -6 \\ 39 \end{pmatrix} \\ &= 289\end{aligned}$$

Example #2: Part (c) continued

Answer for 2(c) continued:

So, the needed probability can be obtained as

$$\begin{aligned}P(4X_1 - 3X_2 + 5X_3 < 63) &= P\left(Z < \frac{63 - 46}{\sqrt{289}}\right) \\&= P(Z < 1) \\&= \Phi(1) \\&= 0.8413447\end{aligned}$$

Example #2: R Code

```
# Example 2a
> pnorm(1.5, lower=F)
[1] 0.0668072
> pnorm(8, mean=5, sd=2, lower=F)
[1] 0.0668072

# Example 2b
> pnorm(2.25/sqrt(119/32), lower=F)
[1] 0.1216523
> pnorm(8, mean=5.75, sd=sqrt(119/32), lower=F)
[1] 0.1216523

# Example 2c
> pnorm(1)
[1] 0.8413447
> pnorm(63, mean=46, sd=17)
[1] 0.8413447
```

Example #2: Python Code

```
from math import sqrt
from scipy.stats import norm
|
# Example 2a
ans_2a_z = norm.sf(1.5) # P(Z > 1.5)
ans_2a_x1 = norm.sf(8, loc=5, scale=2) # P(X1 > 8), X1~N(5,4)
print("2a:", ans_2a_z, ans_2a_x1)

# Example 2b
# R uses: pnorm(2.25/sqrt(119/32), lower=F) and pnorm(8, mean=5.75, sd=sqrt(119/32), lower=F)
sd_cond = sqrt(119/32)
ans_2b_z = norm.sf(2.25 / sd_cond) # P(Z > 2.25/sd)
ans_2b_x1 = norm.sf(8, loc=5.75, scale=sd_cond) # P(X1 > 8 | X2=1, X3=10)
print("2b:", ans_2b_z, ans_2b_x1)

# Example 2c
ans_2c_z = norm.cdf(1) # P(Z < 1)
ans_2c_lin = norm.cdf(63, loc=46, scale=17) # P(4X1-3X2+5X3 < 63) ~ N(46, 289)
print("2c:", ans_2c_z, ans_2c_lin)
```

... 2a: 0.06680720126885807 0.06680720126885807
2b: 0.1216522986681281 0.1216522986681281
2c: 0.8413447460685429 0.8413447460685429



Likelihood Function

Suppose that $\mathbf{x}_i = (x_{i1}, \dots, x_{ip})$ is a sample from a normal distribution with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$, i.e., $\mathbf{x}_i \stackrel{\text{iid}}{\sim} N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

The likelihood function for the parameters (given the data) has the form

$$L(\boldsymbol{\mu}, \boldsymbol{\Sigma} | \mathbf{X}) = \prod_{i=1}^n f(\mathbf{x}_i) = \prod_{i=1}^n \frac{1}{(2\pi)^{p/2} |\boldsymbol{\Sigma}|^{1/2}} \exp \left\{ -\frac{1}{2} (\mathbf{x}_i - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu}) \right\}$$

and the log-likelihood function is given by

$$LL(\boldsymbol{\mu}, \boldsymbol{\Sigma} | \mathbf{X}) = -\frac{np}{2} \log(2\pi) - \frac{n}{2} \log(|\boldsymbol{\Sigma}|) - \frac{1}{2} \sum_{i=1}^n (\mathbf{x}_i - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \boldsymbol{\mu})$$

Maximum Likelihood Estimate of Mean Vector

The MLE of the mean vector is the value of μ that minimizes

$$\sum_{i=1}^n (\mathbf{x}_i - \mu)' \Sigma^{-1} (\mathbf{x}_i - \mu) = \sum_{i=1}^n \mathbf{x}_i' \Sigma^{-1} \mathbf{x}_i - 2n\bar{\mathbf{x}}' \Sigma^{-1} \mu + n\mu' \Sigma^{-1} \mu$$

where $\bar{\mathbf{x}} = (1/n) \sum_{i=1}^n \mathbf{x}_i$ is the sample mean vector.

Taking the derivative with respect to μ we find that

$$\frac{\partial LL(\mu, \Sigma | \mathbf{X})}{\partial \mu} = -2n\Sigma^{-1}\bar{\mathbf{x}} + 2n\Sigma^{-1}\mu \quad \longleftrightarrow \quad \bar{\mathbf{x}} = \hat{\mu}$$

The sample mean vector $\bar{\mathbf{x}}$ is the MLE of the population mean μ vector.

Maximum Likelihood Estimate of Covariance Matrix

The MLE of the covariance matrix is the value of Σ that minimizes

$$-n \log(|\Sigma^{-1}|) + \sum_{i=1}^n \text{tr}\{\Sigma^{-1}(\mathbf{x}_i - \hat{\mu})(\mathbf{x}_i - \hat{\mu})'\}$$

where $\hat{\mu} = \bar{\mathbf{x}} = (1/n) \sum_{i=1}^n \mathbf{x}_i$ is the sample mean.

Taking the derivative with respect to Σ^{-1} we find that

$$\frac{\partial LL(\mu, \Sigma | \mathbf{X})}{\partial \Sigma^{-1}} = -n\Sigma + \sum_{i=1}^n (\mathbf{x}_i - \hat{\mu})(\mathbf{x}_i - \hat{\mu})'$$

i.e., the sample covariance matrix $\hat{\Sigma} = (1/n) \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})'$ is the MLE of the population covariance matrix Σ .

Sampling Distributions

Univariate Sampling Distributions: $\bar{x}$ and s^2

In the univariate normal case, we have that

- $\bar{x} = (1/n) \sum_{i=1}^n x_i \sim N(\mu, \sigma^2/n)$
- $(n-1)s^2 = \sum_{i=1}^n (x_i - \bar{x})^2 \sim \sigma^2 \chi_{n-1}^2$

χ_k^2 denotes a **chi-square variable** with k degrees of freedom.

$$\sigma^2 \chi_k^2 = \sum_{i=1}^k z_i^2 \text{ where } z_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

Multivariate Sampling Distributions: $\bar{\mathbf{x}}$ and $\mathbf{S}$

In the multivariate normal case, we have that

- $\bar{\mathbf{x}} = (1/n) \sum_{i=1}^n \mathbf{x}_i \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma}/n)$
- $(n-1)\mathbf{S} = \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})' \sim W_{n-1}(\boldsymbol{\Sigma})$

$W_k(\boldsymbol{\Sigma})$ denotes a **Wishart variable** with k degrees of freedom.

$$W_k(\boldsymbol{\Sigma}) = \sum_{i=1}^k \mathbf{z}_i \mathbf{z}_i' \text{ where } \mathbf{z}_i \stackrel{\text{iid}}{\sim} N(\mathbf{0}_p, \boldsymbol{\Sigma})$$